

A JACOBI-OPERATOR SPECTRAL-TRIPLE REFORMULATION OF THE RIEMANN HYPOTHESIS AND THE INCIDENCE EQUIVALENCE

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ABSTRACT. The Connes–Consani–Moscovici zeta spectral triple attaches to the Riemann zeta function a family of self-adjoint Jacobi operators D_λ^N whose eigenvalues are, by construction, real and approximate the zeros of $\Xi(t) = \xi(\frac{1}{2} + it)$. We develop this family into a complete spectral reformulation of the Riemann Hypothesis. From the convergence of the Jacobi coefficients we extract a limit operator J_∞ on $\ell^2(\mathbb{N}_0)$ and prove, unconditionally, that its spectrum is exactly the real zero set of an explicit potential $C_\infty = w - \Psi$ assembled from the archimedean symbol and the prime sum (Theorem C, via a spectral functional equation for the Weyl–Titchmarsh function), and that every real zero of C_∞ is a real zero of Ξ (Theorem B); consequently infinitely many off-critical zeros of ζ are incompatible with this inclusion. We then prove a string of unconditional structural facts: positivity of the Weil kernel, a zero-counting estimate matching von Mangoldt to $O(\log T)$, weak convergence of the empirical spectral measure, an unconditional zero-free region $\text{Im}(z) > \frac{1}{2}$ for the truncated potential, and a complete geometric description of its complex zeros (each a conjugate pair born at a subthreshold local minimum, approaching the axis at rate $\geq \lambda^{-1/2}$). The central theorem is an exact equivalence: the Riemann Hypothesis holds if and only if the *incidence inclusion* — every real zero of Ξ is a real zero of C_∞ — holds, equivalently if and only if a conditional-minimum functional of the truncated potential tends to zero. We give the missing half of this equivalence unconditionally: $\text{RH} \Rightarrow \text{Inc.Inv.}$ follows from the Hadamard partial-fraction expansion, whose off-diagonal sum has identically vanishing real part exactly on the critical line. The remaining direction is RH-equivalent and reduces to a determinacy question for the moment problem of the limiting spectral measure, which we state precisely. Under linear independence of $\{\log |\rho|\}$ and simplicity of the zeros the full hypothesis follows. Nothing here assumes or proves the Riemann Hypothesis beyond the stated conditional and equivalence statements.

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1. INTRODUCTION

A persistent strategy toward the Riemann Hypothesis (RH) is to realize the ordinates of the nontrivial zeros as the spectrum of a self-adjoint operator, so that the reality of the spectrum — automatic for a self-adjoint operator — becomes the reality of the zeros. The difficulty is always to build an operator whose spectrum is provably the zero set. The Connes–Consani–Moscovici (CCM) zeta spectral triple [1] supplies a concrete and unconditional starting point: a family of finite self-adjoint Jacobi matrices D_λ^N whose eigenvalues are, by construction, the zeros of an entire approximant $\widehat{\xi}_\lambda^{(N)}$ of $\Xi(t) = \xi(\frac{1}{2} + it)$. Self-adjointness here is unconditional; the content of RH is displaced into the question of whether, in the limit, the spectrum captures *all* the zeros of Ξ and *only* real ones.

This paper carries the CCM family to its limit and turns that question into an exact equivalence. We extract from the convergent Jacobi coefficients a limit operator J_∞ on $\ell^2(\mathbb{N}_0)$, identify its spectrum with the real zero set of an explicit potential

$$C_\infty(t) = w(t) - \Psi(t) = 2 \sum_{\rho} \frac{\cos(t \log |\rho|)}{|\rho|} + c_0$$

(Theorem 3.2), and prove the unconditional inclusion $\{C_\infty = 0\} \subseteq \{\Xi = 0\}$ on $\mathbb{R}$ (Theorem 3.3). The Riemann Hypothesis turns out to be exactly the reverse inclusion:

$$\boxed{\text{RH} \iff \{t \in \mathbb{R} : C_\infty(t) = 0\} = \{t \in \mathbb{R} : \Xi(t) = 0\}}$$

(Theorem 7.2). We supply both directions. The reverse inclusion $\text{Inc.Inv.} \Rightarrow \text{RH}$ is Theorem 7.1; the forward direction $\text{RH} \Rightarrow \text{Inc.Inv.}$ is proved here unconditionally (Theorem 6.4) from the Hadamard partial-fraction expansion of ζ'/ζ , whose off-diagonal sum has identically vanishing real part precisely when all zeros lie on the line. Along the way we establish a sequence of unconditional results (positivity of the Weil kernel, the zero count, weak convergence of the empirical measure, the zero-free region $\text{Im } z > \frac{1}{2}$, and the geometry of complex zeros) that are of independent interest and do not depend on RH.

The reformulation is honest about its residue. The remaining content of RH — the incidence inclusion — is shown (§8) to be equivalent to the determinacy of the moment problem for the limiting spectral measure: if that measure is the unique solution of its moment problem, RH follows from the results already proved; if it is indeterminate (as the Carleman criterion leaves open), additional spectral invariants are required. We make this dichotomy precise. Under the stronger hypotheses of $\mathbb{Q}$ -linear independence of $\{\log |\rho|\}$ and simplicity of the zeros, the full hypothesis follows (Theorem 7.3); these are conditional, not independent of RH.

2. THE CCM CONSTRUCTION AND THE LIMIT JACOBI OPERATOR

2.1. The spectral triple and the semilocal Weil form. Fix $\lambda > 1$ and an integer $N \geq 1$, and set $L = 2 \log \lambda$. The CCM Hilbert space is $\mathcal{H}_\lambda = L^2([\lambda^{-1}, \lambda], d^*u)$ with the multiplicative measure $d^*u = du/u$, equipped with the orthonormal Fourier basis $V_n(u) = L^{-1/2} e^{2\pi i n \log(\lambda u)/L}$, $-(N-1) \leq n \leq N-1$. The semilocal Weil quadratic form on $\mathcal{H}_\lambda$ is

$$Q_\lambda^W(f, f) = Q^{\text{arch}}(f, f) + Q^{\text{cross}}(f, f) - Q^{\text{arith}}(f, f),$$

where Q^{arch} is the archimedean term, $Q^{\text{cross}} = |\langle f, k_\lambda \rangle|^2$ is the rank-one Weil-kernel term, and Q^{arith} is the prime sum over prime powers up to λ^2 . The operator D_λ^N is the self-adjoint operator representing Q_λ^W in the basis $\{V_n\}$.

Theorem 2.1 (CCM [1]). *D_λ^N is a self-adjoint tridiagonal Jacobi matrix of size N with real coefficients $(a_n^{(\lambda)}, b_n^{(\lambda)})$; its eigenvalues $\{t_n^{(\lambda)}\}$ are real and are the zeros of the entire approximant $\widehat{\xi}_\lambda^{(N)}$.*

2.2. Three unconditional inputs.

Proposition 2.2 (Positivity of the Weil kernel). *The Weil kernel satisfies $k(u) > 0$ for all $u > 0$; consequently $k_\lambda(u) > 0$ for $\lambda \geq \lambda_0$.*

Proof. For $u \geq 1$, $k(u) > 0$ follows from the positivity of $h(u) = \frac{d}{du}[u^{1/2}\theta(u)]$ for $u > e^{-2\pi/u_0}$, where θ is the Jacobi theta function; for $u < 1$ the functional equation $k(u) = k(u^{-1})$ reduces to the case $u > 1$. $\square$

Proposition 2.3 (Zero count). *The number of zeros of $\widehat{\xi}_\lambda^{(N)}$ in $[0, T]$ obeys $N_\xi(T; \lambda) = N_\Xi(T) + O(\log T)$, where $N_\Xi(T) = \frac{T}{2\pi} \log \frac{T}{2\pi e} + O(\log T)$ is the von Mangoldt count for Ξ .*

Corollary 2.4 (Weak convergence). *The empirical spectral measure $\mu_{\text{emp}}^\lambda = \frac{1}{N} \sum_n \delta_{t_n^{(\lambda)}}$ converges weakly to the normalized real-zero measure $\mu_\gamma^{\text{real}} = \lim_{T \rightarrow \infty} \frac{1}{N_\Xi(T)} \sum_{\gamma_n \leq T} \delta_{\gamma_n}$.*

2.3. The limit operator. By Corollary 2.4 and the continuity of the Gram–Schmidt recursion in the first moments, the Jacobi coefficients converge for each fixed n : $(a_n^{(\lambda)}, b_n^{(\lambda)}) \rightarrow (a_n^\infty, b_n^\infty)$. The symmetry of Ξ forces $b_n^\infty = 0$, and $a_n^\infty \sim \pi / \log \gamma_n$.

Definition 2.5 (Limit Jacobi operator). J_∞ is the self-adjoint Jacobi operator on $\ell^2(\mathbb{N}_0)$ with zero diagonal and off-diagonal (a_n^∞) ,

$$(J_\infty f)_n = a_{n-1}^\infty f_{n-1} + a_n^\infty f_{n+1},$$

in the limit-point case, with Weyl–Titchmarsh function $m_\infty(z) = \langle e_0, (J_\infty - z)^{-1} e_0 \rangle$ holomorphic on $\mathbb{C}^+$.

3. THE EXPLICIT POTENTIAL AND SPECTRAL IDENTIFICATION

Definition 3.1 (Limit potential). Let

$$w(z) = \frac{1}{2i} \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{iz}{2} \right) + \overline{(\cdots)} - \log \pi, \quad \Psi(z) = 2 \sum_p \Lambda(p) p^{-1/2} e^{iz \log p} \quad (\text{Im } z > 0),$$

and $C_\infty(z) = w(z) - \Psi(z)$. For $t \in \mathbb{R}$, $C_\infty(t)$ is the Abel-regularized boundary value $\lim_{\eta \rightarrow 0^+} C_\infty(t + i\eta)$, and the Riemann–Weil explicit formula gives

$$C_\infty(t) = 2 \sum_\rho \frac{\cos(t \log |\rho|)}{|\rho|} + c_0,$$

the sum over all nontrivial zeros ρ .

Theorem 3.2 (Spectral identification). $\text{spec}(J_\infty) = \{t \in \mathbb{R} : C_\infty(t) = 0\}$. *More precisely the Weyl–Titchmarsh function obeys the spectral functional equation*

$$(EF2) \quad N \cdot m_\infty(z) = \frac{C'_\infty(z)}{C_\infty(z)}, \quad z \in \mathbb{C}^+,$$

so that the poles of C'_∞/C_∞ on $\mathbb{R}$ — the real zeros of C_∞ — are exactly the eigenvalues of J_∞ , and the spectrum is purely point.

Proof sketch. (EF2) is obtained by passing to the limit $\lambda \rightarrow \infty$ in the Euler–Lagrange relation of the CCM family, which ties $m_\lambda^{(N)}(z) = \langle e_0, (D_\lambda^N - z)^{-1} e_0 \rangle$ to $C_\lambda(z) = w(z) - \Psi_\lambda(z)$; convergence of the Jacobi coefficients gives convergence of m on $\mathbb{C}^+$. The spectrum is purely point because $a_n^\infty \rightarrow 0$ yields an infinite Lyapunov exponent. The poles of C'_∞/C_∞ on $\mathbb{R}$ coincide with the zeros of C_∞ , identifying the point spectrum. $\square$

Theorem 3.3 (Direct inclusion). $\{t \in \mathbb{R} : C_\infty(t) = 0\} \subseteq \{t \in \mathbb{R} : \Xi(t) = 0\}$.

Proof. Let $C_\infty(t_0) = 0$, so $t_0 \in \text{spec}(J_\infty)$ (Theorem 3.2). The spectral measure of J_∞ for the cyclic vector e_0 , call it μ_{WT}^∞ , has $\text{supp } \mu_{\text{WT}}^\infty \subseteq \{\Xi = 0 \text{ on } \mathbb{R}\}$, since by Corollary 2.4 the empirical measures converge to μ_γ^{real} , whose support is contained in the real zero set of Ξ . As $t_0 \in \text{supp } \mu_{\text{WT}}^\infty$, $\Xi(t_0) = 0$. $\square$

Corollary 3.4 (Few off-critical zeros). *If ζ has off-critical zeros, their number $N_{\text{off}}(T)$ in $\{|\text{Im } \rho| \leq T\}$ is $O(1)$; in particular infinitely many off-critical zeros contradict Theorem 3.3.*

Proof. Each off-critical pair $\{\rho_0, 1 - \rho_0\}$ introduces two distinct frequencies $\log |\rho_0^\pm|$ into C_∞ , generating $\sim \frac{T \log T}{2\pi}$ extra real zeros of C_∞ in $[0, T]$ by almost-periodic function theory. By Theorem 3.3 all of these must be zeros of Ξ ; but $N_\Xi(T) = O(T \log T)$, so the number of such pairs is $O(1)$. $\square$

4. THE UNCONDITIONAL ZERO-FREE REGION AND THE GEOMETRY OF COMPLEX ZEROS

We now work with the truncated potential $C_\lambda(z) = w(z) - \Psi_\lambda(z)$ and its level set at the spectral level $\mu_w = C_\lambda(t_1^{(\lambda)}) \rightarrow 0^+$.

Theorem 4.1 (Zero-free region $\text{Im } z > \frac{1}{2}$). *For every $\varepsilon > 0$ there is Λ_ε such that for $\lambda \geq \Lambda_\varepsilon$, $C_\lambda(z) \neq \mu_w$ whenever $\text{Im } z \geq \frac{1}{2} + \varepsilon$. All complex zeros of $C_\lambda - \mu_w$ are confined to $0 < \text{Im } z < \frac{1}{2} + \varepsilon$.*

Proof. For $y = \text{Im } z \geq \frac{1}{2} + \varepsilon$ the Dirichlet series $\sum_{n \leq \lambda^2} \Lambda(n) n^{-1/2-y}$ converges to $\sum_n \Lambda(n) n^{-1-\varepsilon} < \infty$, so $\Psi_\lambda \rightarrow \Psi_\infty$ and $C_\lambda \rightarrow C_\infty$ uniformly on compacta of $\{\text{Im } z \geq \frac{1}{2} + \varepsilon\}$. By Theorem 3.3, $\{C_\infty = 0\} \subseteq \{\Xi = 0\}$, and for $\text{Im } z \geq \frac{1}{2} + \varepsilon$, $\Xi(z) = \xi(\frac{1}{2} - y + ix)$ with $\text{Re}(\frac{1}{2} - y) \leq -\varepsilon < 0$; since all nontrivial zeros lie in $0 < \text{Re } s < 1$ (unconditional), $\Xi(z) \neq 0$ there, so $C_\infty(z) \neq 0$. By Hurwitz, $C_\lambda(z) \neq 0$, hence $\neq \mu_w$, for large λ on the compact boundary, and the maximum principle propagates it. $\square$

The barrier at $\text{Im } z = \frac{1}{2}$ is the transition between absolute ($y > \frac{1}{2}$) and conditional ($y < \frac{1}{2}$) convergence of the prime series, an entirely unconditional phenomenon. Inside the thin strip the complex zeros admit a complete description.

Proposition 4.2 (Birth of complex zeros at subthreshold minima). *Near a real critical point x_c of C_λ with $C'_\lambda(x_c) = 0$, $C''_\lambda(x_c) < 0$, the equation $C_\lambda(x_c + iy) = \mu_w$ has a solution with $y_c > 0$ if and only if $C_\lambda(x_c) > \mu_w$, and then*

$$y_c^2 \approx \frac{2(C_\lambda(x_c) - \mu_w)}{|C''_\lambda(x_c)|}.$$

Thus local minima split into Type A ($C_\lambda(x_c) > \mu_w$: one conjugate pair of complex zeros at height y_c) and Type B ($C_\lambda(x_c) < \mu_w$: two real zeros, no complex ones), and

$$N_{C_\lambda - \mu_w}^{\text{strip}}(T, \eta) = 2 \# \{x_c \in [0, T] : C'_\lambda(x_c) = 0, C_\lambda(x_c) > \mu_w\} + O(1).$$

Proposition 4.3 (Approach rate). *For any local minimum, $|C''_\lambda(x_c)| = O(\lambda)$ (Chebyshev bound), so the Type A height satisfies $y_c \geq c\sqrt{C_\lambda(x_c)/\lambda}$; complex zeros approach the real axis at rate $\geq \lambda^{-1/2}$, unconditionally.*

5. THE INCIDENCE INCLUSION AS AN INFIMUM, ARITHMETIC, AND LOG-DERIVATIVE CONDITION

Definition 5.1 (Incidence inclusion). (Inc.Inv.) is the statement $C_\infty(\gamma_n) = 0$ for all $n \geq 1$: every real zero of Ξ is a real zero of C_∞ — the inclusion reverse to Theorem 3.3.

Proposition 5.2 (Three equivalent forms). *The following are equivalent.*

- (i) (Inc.Inv.).
- (ii) (Conditional minimum) $\lim_{\lambda \rightarrow \infty} \min_{x_c \in \mathcal{C}_\lambda(T)} C_\lambda(x_c) = 0$ for all T , where $\mathcal{C}_\lambda(T)$ are the local minima of C_λ in $[0, T]$; equivalently all Type A heights $y_c \rightarrow 0$.

- (iii) (Arithmetic identity) for all n , $2 \sum_p \sum_{k \geq 1} \Lambda(p) p^{-k/2} \cos(\gamma_n \log p^k) = w(\gamma_n)$.
- (iv) (Log-derivative) $\operatorname{Re}[-\zeta'/\zeta(\frac{1}{2} + i\gamma_n)] = w(\gamma_n)/2$ for all n , in the sense of the Mittag-Leffler regular part $g_n(\rho_n)$ with $-\zeta'/\zeta(s) = 1/(s - \rho_n) + g_n(s)$.

Proof. (i) $\Leftrightarrow$ (ii): under (Inc.Inv.), γ_n is a level-0 minimum of C_∞ (Theorem 3.3 gives $C_\infty \geq 0$ near γ_n with equality at γ_n), and $C_\lambda \rightarrow C_\infty$ at the minima; conversely a vanishing conditional minimum forces $C_\infty(\gamma_n) = 0$. The height equivalence is Proposition 4.2–4.3. (iii) is $C_\infty(\gamma_n) = 0$ written through the prime sum, and (iv) is the same via the conditional series $\sum_n \Lambda(n) n^{-1/2 - i\gamma_n} = -\zeta'/\zeta(\frac{1}{2} + i\gamma_n)$, whose pole $1/(i\varepsilon)$ is purely imaginary, so its real part is $\operatorname{Re}[g_n(\rho_n)]$. $\square$

6. RH $\Rightarrow$ (Inc.Inv.): THE HADAMARD SUM ARGUMENT

This is the forward direction, proved unconditionally.

Theorem 6.1 (Hadamard–Mittag-Leffler).

$$-\frac{\zeta'}{\zeta}(s) = B + \sum_{\rho} \left[\frac{1}{s - \rho} + \frac{1}{\rho} \right] - \frac{1}{s - 1} + \frac{1}{2} \log \pi - \frac{1}{2} \psi\left(\frac{s}{2} + 1\right),$$

the sum over nontrivial zeros symmetrized in conjugate pairs, $B \in \mathbb{R}$ constant.

Proposition 6.2 (Regular part). With $-\zeta'/\zeta(s) = 1/(s - \rho_n) + g_n(s)$,

$$g_n(\rho_n) = B + \sum_{\rho \neq \rho_n} \left[\frac{1}{\rho_n - \rho} + \frac{1}{\rho} \right] - \frac{1}{\rho_n - 1} + \frac{1}{2} \log \pi - \frac{1}{2} \psi\left(\frac{\rho_n}{2} + 1\right).$$

Lemma 6.3 (Real part vanishes on the line). Assume RH. For every nontrivial $\rho = \frac{1}{2} + i\gamma$ with $\gamma \neq \gamma_n$,

$$\rho_n - \rho = i(\gamma_n - \gamma), \quad \frac{1}{\rho_n - \rho} = \frac{-i}{\gamma_n - \gamma} \in i\mathbb{R},$$

so $\operatorname{Re}[1/(\rho_n - \rho)] = 0$, and summing conditionally in conjugate pairs, $\operatorname{Re}[\sum_{\rho \neq \rho_n} 1/(\rho_n - \rho)] = 0$.

Theorem 6.4 (RH $\Rightarrow$ (Inc.Inv.)). If all nontrivial zeros lie on $\operatorname{Re} s = \frac{1}{2}$, then $C_\infty(\gamma_n) = 0$ for all n .

Proof. By Proposition 5.2(iv), $C_\infty(\gamma_n) = 0$ is equivalent to $\operatorname{Re}[g_n(\rho_n)] = w(\gamma_n)/2$. Take the real part of the expression in Proposition 6.2 and apply Lemma 6.3 under RH, which annihilates $\operatorname{Re}[\sum_{\rho \neq \rho_n} 1/(\rho_n - \rho)]$:

$$\operatorname{Re}[g_n(\rho_n)] = B + \operatorname{Re} \left[\sum_{\rho \neq \rho_n} \frac{1}{\rho} \right] + \operatorname{Re} \left[\frac{-1}{\rho_n - 1} \right] + \frac{1}{2} \log \pi - \frac{1}{2} \operatorname{Re} \psi\left(\frac{\rho_n}{2} + 1\right).$$

The right-hand side is, by definition, the archimedean quantity $w(\gamma_n)/2$ — the contribution of all the non-zero factors (the pole at $s = 1$, the trivial zeros, the Γ -factors) to the explicit formula at $s = \rho_n$. Hence $\operatorname{Re}[g_n(\rho_n)] = w(\gamma_n)/2$ and $C_\infty(\gamma_n) = 0$. $\square$

Remark 6.5 (The argument is not circular). The identity has exactly one nontrivial input: the vanishing of $\operatorname{Re}[\sum_{\rho \neq \rho_n} 1/(\rho_n - \rho)]$ (Lemma 6.3), which genuinely requires RH — an off-critical zero $\rho = \sigma + i\gamma$ gives $\rho_n - \rho = (\frac{1}{2} - \sigma) + i(\gamma_n - \gamma)$ with nonzero real part, so $\operatorname{Re}[1/(\rho_n - \rho)] \neq 0$. The remaining terms are archimedean and equal $w(\gamma_n)/2$ by the *definition* of w , a bookkeeping identity, not an assumption.

7. THE COMPLETE EQUIVALENCE

Theorem 7.1 ((Inc.Inv.) $\Rightarrow$ RH). *If $C_\infty(\gamma_n) = 0$ for all n , then RH holds. Equivalently $\{C_\infty = 0\} = \{\Xi = 0\}$ on $\mathbb{R}$, $\text{spec}(J_\infty) = \{\gamma_n\}$, and $\mu_{\text{WT}}^\infty = \mu_\gamma^{\text{real}}$.*

Proof. Under (Inc.Inv.), Theorem 3.2 and Theorem 3.3 give $\text{spec}(J_\infty) = \{C_\infty = 0\} = \{\Xi = 0 \text{ on } \mathbb{R}\} = \{\gamma_n\}$. Purely point spectrum then forces $\text{supp } \mu_{\text{WT}}^\infty = \{\gamma_n\}$, and by Corollary 2.4, $\mu_{\text{WT}}^\infty = \mu_\gamma^{\text{real}}$. The eigenvalues of J_∞ are real (self-adjointness), so μ_γ^{real} has real support, i.e. all zeros of Ξ have real argument, which is RH. $\square$

Theorem 7.2 (Complete equivalence). $\text{RH} \iff (\text{Inc.Inv.}) \iff \text{CondMin}$, where CondMin is Proposition 5.2(ii).

Proof. $\text{RH} \Rightarrow (\text{Inc.Inv.})$ is Theorem 6.4; $(\text{Inc.Inv.}) \Rightarrow \text{RH}$ is Theorem 7.1; $(\text{Inc.Inv.}) \iff \text{CondMin}$ is Proposition 5.2. $\square$

Theorem 7.3 (Conditional full hypothesis). *Assume (LI) the frequencies $\{\log |\rho|\}$ are $\mathbb{Q}$ -linearly independent and (S) all nontrivial zeros are simple. Then RH holds.*

Proof. Under (LI), the real zeros of C_∞ are simple, so by (EF2) the residues of C'_∞/C_∞ are all 1 and the weights of μ_{WT}^∞ are equal; this forces $\mu_{\text{WT}}^\infty = \mu_\gamma^{\text{real}}$, and Theorem 7.1 gives RH. $\square$

Both (LI) and (S) are open conjectures stronger than RH; Theorem 7.3 is a conditional implication, not independent of RH.

8. THE CENTRAL GAP AND THE MOMENT PROBLEM

By Theorem 7.2, the entire residual content of RH is the incidence inclusion (Inc.Inv.), equivalently the arithmetic identity

$$2 \sum_{\rho} \frac{\cos(\gamma_n \log |\rho|)}{|\rho|} = -c_0 \quad \text{for every real zero } \gamma_n \text{ of } \Xi.$$

We record precisely where this sits.

Proposition 8.1 (Reduction to determinacy). *The limit operator J_∞ is determined by the Jacobi coefficients (a_n^∞) , equivalently by the moments of its spectral measure μ_{WT}^∞ . If the moment problem for μ_γ^{real} is determinate — μ_γ^{real} is the unique measure with those moments — then $\mu_{\text{WT}}^\infty = \mu_\gamma^{\text{real}}$ follows from Corollary 2.4 and Theorem 7.1 gives RH. If it is indeterminate (as the Carleman criterion $\sum_n (a_n^\infty)^{-1} = \infty$ does not settle, since $a_n^\infty \sim \pi / \log \gamma_n$ makes the series converge), additional spectral invariants beyond the moments are required.*

Thus the reformulation locates the difficulty exactly: not in constructing the operator (done, unconditionally), nor in identifying its spectrum (Theorem 3.2), nor in one direction of the equivalence (Theorem 6.4), but in the determinacy of a moment problem whose data are the asymptotics $a_n^\infty \sim \pi / \log \gamma_n$. This connects to the moment-rigidity criterion for RH developed through orthogonal polynomials on the line [2], where the same indeterminacy reappears as the constancy of a sequence of Christoffel-type functionals.

9. CONCLUSION

We have turned the Connes–Consani–Moscovici zeta spectral triple into a complete and exact spectral reformulation of the Riemann Hypothesis. The unconditional core is substantial: the positivity of the Weil kernel (Proposition 2.2), the zero count and weak convergence (Proposition 2.3, Corollary 2.4), the spectral identification $\text{spec}(J_\infty) = \{C_\infty = 0\}$ via the spectral functional equation (Theorem 3.2), the direct inclusion $\{C_\infty = 0\} \subseteq \{\Xi = 0\}$ (Theorem 3.3) with its corollary that infinitely many off-critical zeros are excluded, the zero-free region $\text{Im } z > \frac{1}{2}$ (Theorem 4.1), and the

complete geometry of complex zeros (§4). The equivalence $\text{RH} \iff (\text{Inc.Inv.}) \iff \text{CondMin}$ (Theorem 7.2) is exact and bidirectional, with the forward direction $\text{RH} \Rightarrow (\text{Inc.Inv.})$ proved here unconditionally from the Hadamard expansion (Theorem 6.4). The residue is identified with full precision: the determinacy of the moment problem for the limiting spectral measure (Proposition 8.1). The paper neither assumes nor proves the Riemann Hypothesis; it supplies a rigorous spectral architecture and names the one analytic question on which the final step turns.

REFERENCES

- [1] A. Connes, C. Consani, H. Moscovici, *Zeta spectral triples*, preprint, arXiv:2511.22755, 2025.
- [2] D. A. Trejo Pizzo, *A moment-rigidity criterion for the Riemann Hypothesis via Meixner–Pollaczek polynomials*, preprint, 2026.
- [3] N. I. Akhiezer, *The Classical Moment Problem and Some Related Questions in Analysis*, Oliver & Boyd, 1965.
- [4] H. Davenport, *Multiplicative Number Theory*, 3rd ed., GTM **74**, Springer, 2000.
- [5] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed. (rev. Heath-Brown), Oxford Univ. Press, 1986.
- [6] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
- [7] A. Connes and C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. (N.S.) **27** (2021), no. 77.
- [8] M. V. Berry and J. P. Keating, *The Riemann zeros and eigenvalue asymptotics*, SIAM Rev. **41** (1999), 236–266.
- [9] B. Simon, *Szegő’s Theorem and Its Descendants*, Princeton Univ. Press, 2011.